

Week 5 Recitation

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ECON2001.01 (Prin. Micro.)

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Buyer's Problem

Continued

3 Ingredients of Buyer's problem

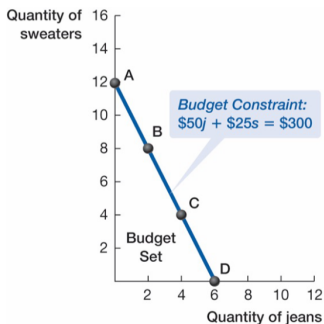
- What you like (=Preference, or Tastes)
- Prices
- How much money you have to spend (=Income, or Wealth)

Same Story:

You like jeans and sweaters. You are going to go shopping for jeans and sweaters. A pair of jeans is \$50, and a sweater is \$25. You have \$300 of money in total.

Buyer's Problem

Revisit Week4 - Budget Set and Budget Constraint



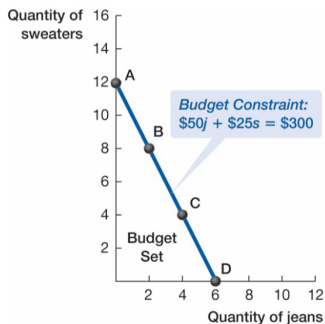
- Opportunity Cost of jeans
 $= \frac{\text{Loss in sweaters}}{\text{Gain in jeans}}$
- Opportunity Cost of sweater
 $= \frac{\text{Loss in jeans}}{\text{Gain in sweaters}}$
- Point A? B? C? D?

The Slope of the Budget constraint

- $|\text{Slope}| = \left| \frac{\text{Vertical Change}}{\text{Horizontal Change}} \right|$
 $= \text{Opportunity Cost of } \underline{\hspace{2cm}}$
- $p_1 x_1 + p_2 x_2 = I$
 $\Leftrightarrow$

Buyer's Problem

Revisit Week4 - Budget Set and Budget Constraint



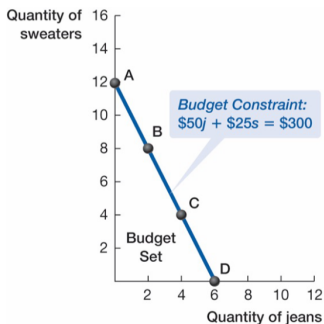
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Buyer's Problem

Revisit Week4 - Budget Set and Budget Constraint



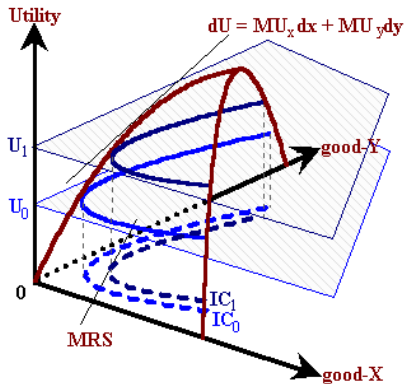
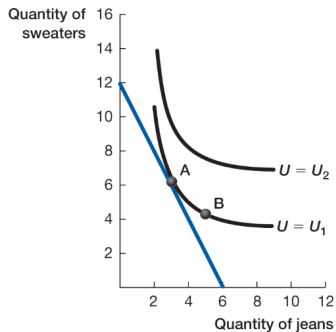
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- ★★★ Slope =

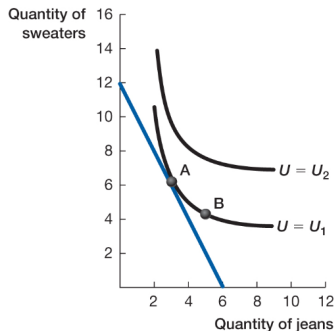
Buyer's Problem

Revisit Week4 - Indifference Curve



Buyer's Problem

Revisit Week4 - Indifference Curve



- **Indifference Curves:**

All possible combinations of good 1 and good 2 that provide the buyer with the same level of utility

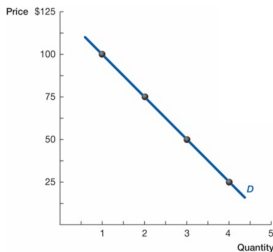
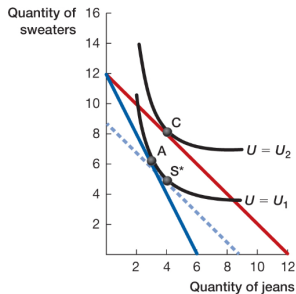
- As moving out from the origin, increasing in utility (benefit)

- Never crosses

- Slope = $-\frac{MB_{Horizontal}}{MB_{Vertical}} = -\frac{MB_1}{MB_2}$

Buyer's Problem

Revisit Week4 - Connecting to Demand

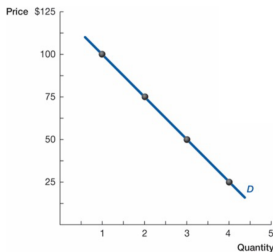
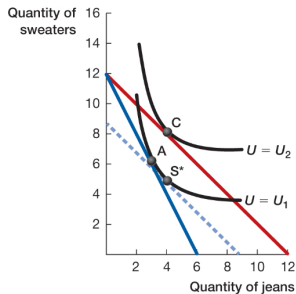


Price	Quantity Demanded
\$25	4 pairs of jeans
50	3 pairs of jeans
75	2 pairs of jeans
100	1 pair of jeans

- Slope of Indifference Curve = Slope of Budget Constraint
- $-\frac{MB_1}{MB_2} =$

Buyer's Problem

Revisit Week4 - Connecting to Demand

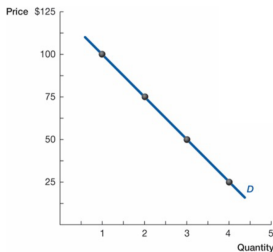
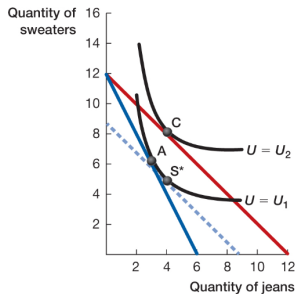


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Buyer's Problem

Revisit Week4 - Connecting to Demand

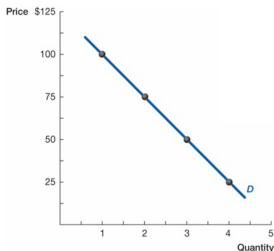
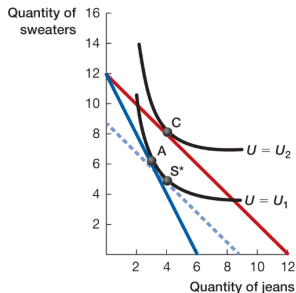


Price	Quantity Demanded
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- Slope of Indifference Curve = Slope of Budget Constraint
- $-\frac{MB_1}{MB_2} = -\frac{p_1}{p_2}$ or $\frac{MB_1}{p_1} = \frac{MB_2}{p_2}$
- Marginal Benefits per Dollar spent
- The “Bang for your buck”

Buyer's Problem

Price Change and Q-demanded



Price	Quantity Demanded
\$25	4 pairs of jeans
50	3 pairs of jeans
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Buyer's Problem

Price Change and Elasticity

Elasticity: A measure of how sensitive one variable is to changes in another

Price elasticity of demand: “How much does Q-demanded change when the good's price changes?”

- Price elasticity of demand = $\frac{\text{Percentage change in}}{\text{Percentage change in}}$

Buyer's Problem

Price Change and Elasticity

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Price elasticity of demand: “How much does Q-demanded change when the good’s price changes?”

- Price elasticity of demand = $\frac{\text{Percentage change in Q-demanded}}{\text{Percentage change in price}}$

Arc elasticity of demand =

$$\underbrace{\left[\underbrace{(Q_2 - Q_1)}_{(\Delta Q)} / \underbrace{(Q_2 + Q_1)/2}_{Q_{\text{avg}}} \right] / \left[\underbrace{(P_2 - P_1)}_{(\Delta P)} / \underbrace{(P_2 + P_1)/2}_{P_{\text{avg}}} \right]}$$

We can begin our calculation with this shortcut version of the arc elasticity equation:

$$(\Delta Q / Q_{\text{avg}}) / (\Delta P / P_{\text{avg}})$$

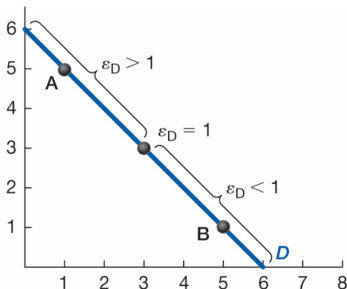
Buyer's Problem

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- Slope is **the same** over the entire linear demand curve (because demand is linear), but elasticity **varies** over ranges of the demand curve.



Buyer's Problem

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Cross-Price Elasticity of demand: “How much does the Q-demanded of one good change when the price of another good changes?”

- Price elasticity of demand = $\frac{\text{Percentage change in Q-demanded of good x}}{\text{Percentage change in price of good y}}$

Buyer's Problem

Price Change and Elasticity

Example:

8. Consider the following demand schedule:

Price	Quantity
\$9	52
\$11	48
\$13	40

- Use the midpoint formula to calculate the elasticity of demand (i) when the price rises from \$9 to \$11 and (ii) when the price rises from \$11 to \$13.
- When the price rises from \$9 to \$11, does expenditure rise, fall, or remain constant? What about when the price rises from \$11 to \$13?

Buyer's Problem

Price Change and Elasticity

Example:

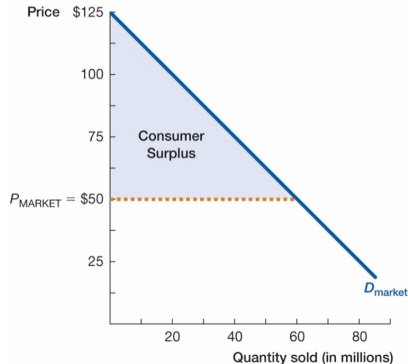
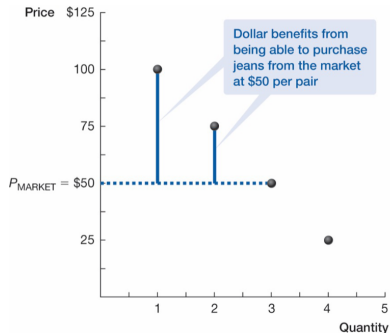
9. You work at a convenience store and make note of changes to sales after the price of liquid soap unexpectedly increases by 10 percent. You notice that sale of shampoo goes *up* by 5 percent and sale of lotion goes *down* by 2 percent. Calculate the cross-price elasticities, and comment on whether each product is a substitute for liquid soap or a complement for liquid soap.

Buyer's Problem

Consumer Surplus

- Consumer Surplus:**

The difference between what a buyer is _____ for a good and what the buyer _____.

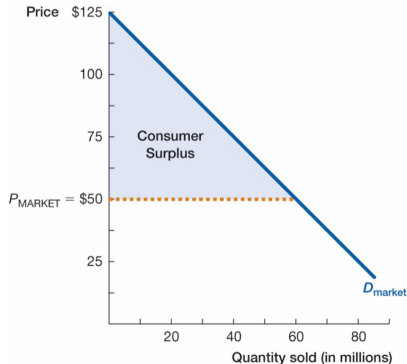
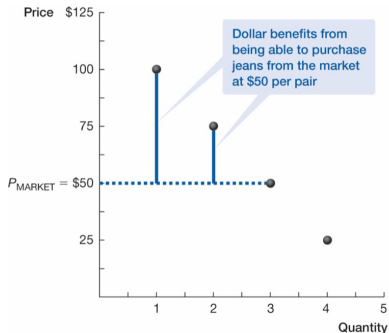


Buyer's Problem

Consumer Surplus

- **Consumer Surplus:**

The difference between what a buyer is willing to pay for a good and what the buyer actually pays.



Buyer's Problem

Consumer Surplus

Example:

13. Suppose demand is given by $Q_D = 6 - P$.
- Graph the demand curve.
 - If the price is \$2, what is the consumer surplus?
 - If the price goes up to \$4, what is the new consumer surplus? This new surplus should be *lower*; explain why.
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